Proposal for Emoji: White Blood Cell

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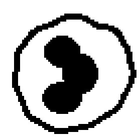
1. Identification

Keywords: leukocyte; immunity; infection; inflammation; laboratory
Category: People & Body body-parts

2. Images

	18x18	72x72
Colour		
Black and white		

The image uses the standard neutrophil form - a pale, irregular cell body with one connected, multi-lobed nucleus - as the visual model for the broader white-blood-cell category. Neutrophils are the most numerous circulating leukocytes, and these cues remain legible at actual keyboard size while separating the image from a spiked microbe, a teardrop-shaped Drop of Blood, and a generic cell diagram.

I, Shuhan He, certify that these example images are original Medical Emoji artwork, that I own or control all IP Rights in them, and that they contain no third-party artwork, logo, trademark, text, digit, or barcode. I release the images under the CC0 1.0 Universal Public Domain Dedication and grant the rights required by the Unicode Emoji Proposal Agreement and License.

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3. Factors for inclusion

a. Multiple meanings

Not applicable.

b. Use in sequences

White Blood Cell can combine with existing emoji to express clear messages without standardized sequences:

- White Blood Cell + Microbe: immune response to infection.
- White Blood Cell + up arrow or down arrow: high or low white-cell count.
- White Blood Cell + Test Tube: blood count, laboratory testing, or research.

c. Breaks new ground

Yes. The current Unicode emoji vocabulary has no human immune cell. Microbe represents a microscopic organism rather than the body's defense cell; Drop of Blood represents blood as a fluid; and Test Tube represents a container or laboratory activity. White Blood Cell adds the missing host immune cell, so a user can distinguish the body's defense cell from the microbe it is responding to.

Current Unicode Emoji List

d. Distinctiveness

White Blood Cell uses a neutrophil as its visual model: a pale, irregular cell body and one bold, connected, three-lobed nucleus. NCBI describes neutrophils as 50% to 70% of circulating leukocytes and their nuclei as three to five connected segments. At 18 pixels, the enclosed lobed nucleus distinguishes the image from the spiked outline of Microbe, the teardrop silhouette of Drop of Blood, the separate circles of Bubbles, and a generic cell diagram. The image does not assert that every leukocyte has neutrophil morphology; it uses the most numerous type as the visible exemplar for the broader category. Vendors may vary the cell outline, lobe count, and internal shading while keeping a pale irregular body and one connected lobed nucleus.

NCBI Bookshelf: Histology, White Blood Cell

The colour and black-and-white examples on page 1 show that these visible cues remain clear at both required sizes and do not depend on colour alone. All artwork displayed in this proposal as proposed emoji artwork is the original work identified in Section 2.

e. Usage level

The clearest uses are talking about the body's immune response and communicating a high, low, or monitored white blood cell count. The required five-source evidence below shows current Search and Video results, recurring Web and Image search interest, and durable published-book usage. Its relative Trends and Ngram levels are below `elephant`, while the term remains present across the modern record.

The National Cancer Institute defines a white blood cell as a blood cell found in blood and lymph tissue that forms part of the immune system and helps fight infection and other diseases. MedlinePlus documents white blood count as a routine test used to diagnose and monitor conditions affecting white blood cells. Search and Video figures below are indexed-result estimates, not counts of individual users or direct measures of emoji demand.

The concept also has documented use in digital health standards. The active LOINC 6690-2 term standardizes automated leukocyte counts, appears in multiple complete-blood-count panels, and supplies language variants for international implementations. The HL7 US Core Implementation Guide publishes a laboratory-result example for a White Blood Cell count using that LOINC code. These sources demonstrate routine interchange of the concept in digital records; they are usage evidence, not a pictographic compatibility claim.

f. Completeness

Not applicable.

g. Compatibility

Not applicable.

4. Counterarguments to factors for exclusion

a. Already represented

No existing emoji or sequence identifies a white blood cell. Microbe + Shield can suggest protection from a germ, but it cannot express the host immune cell or the standardized white-cell-count observation used in laboratory records. Drop of Blood, Test Tube, Microscope, and Hospital provide context without supplying the missing concept.

b. Overly specific

White Blood Cell is the broad leukocyte category, not a particular subtype, disease, procedure, specialty, treatment, or campaign. The representative artwork uses neutrophil morphology because neutrophils are the most numerous circulating leukocytes and their connected lobed nucleus supplies a stable small-size cue. The character name and intended meaning remain White Blood Cell, covering the five major types counted together in the routine white blood count described by MedlinePlus.

c. Open-ended

Red Blood Cell, Platelet, a generic Cell, and individual leukocyte subtypes are the most likely follow-on candidates. White Blood Cell has a specific boundary: it is the upper-level immune-cell concept, the host-cell counterpart to Microbe, and a named count exchanged in LOINC- and HL7-based laboratory records. Individual leukocyte types are narrower members of this proposed category, while a generic Cell would not convey immune defense or a white-cell count. Red Blood Cell and Platelet would present different semantic and substitute questions and are not implied by this proposal. Selection of White Blood Cell therefore stops at the broad leukocyte category rather than beginning a series of blood components or human cells.

d. Transient

White blood cells are a durable biological and medical concept. The Google Books record below extends through 2022 and shows increasing modern published use, while the current Search and Trends evidence documents ongoing use across web, image, and video formats.

e. Faulty comparison

The `elephant` comparisons measure relative frequency only; they do not claim that another emoji creates a precedent. White Blood Cell addresses the specific gap between the body's immune cell and the Microbe it responds to.

5. Other information

Vendors should use a neutrophil-like form as the visual model for the broader White Blood Cell meaning: a pale or neutral irregular cell body and one bold, connected nucleus with three to five lobes. Fine microscopic detail is optional and should yield to legibility at 18x18. Radiating spikes should be avoided because they imply Microbe, and the nucleus should remain a single connected feature rather than a face-like arrangement. The character name should remain White Blood Cell rather than a leukocyte subtype.

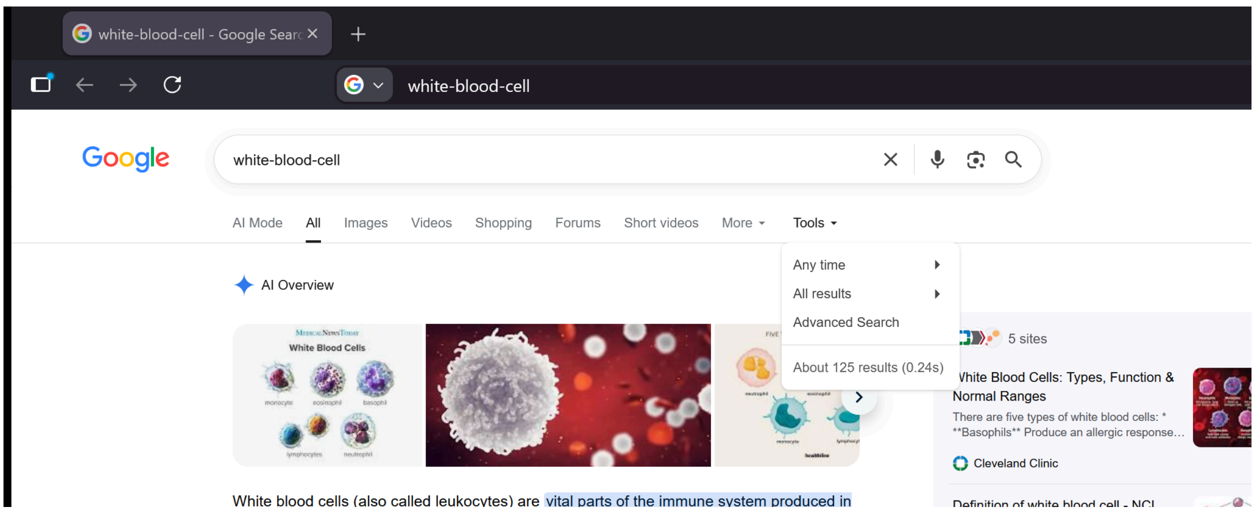
6. Evidence of frequency

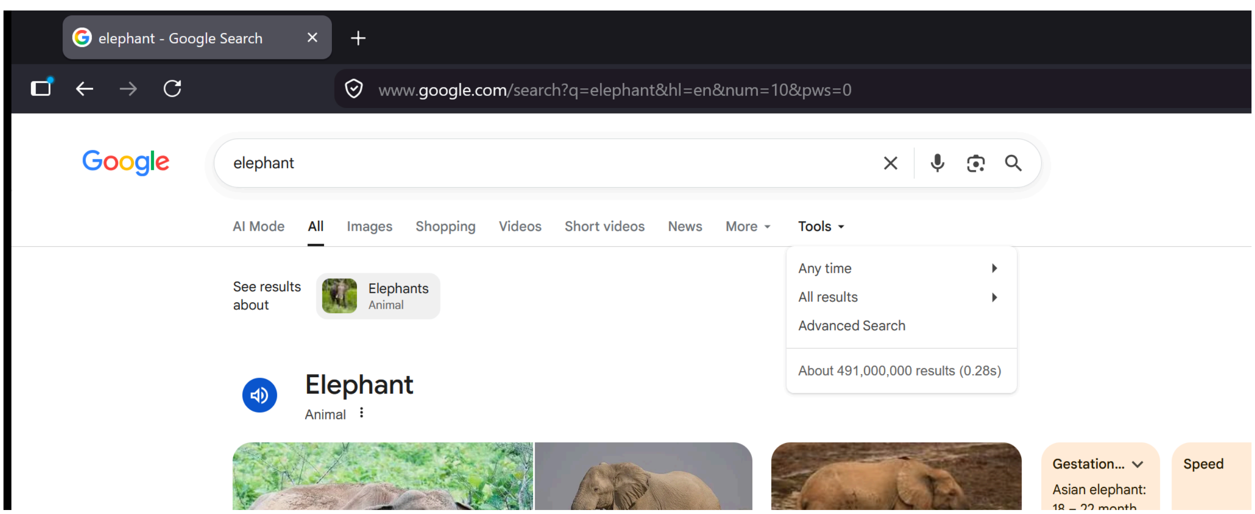
All data are from July 2026 and were obtained in a Firefox Private Browsing window. Search queries disable personalized results. Trends use Worldwide, all categories, and the full available period. Books uses the English corpus, 1500-2022, and smoothing 3.

Google Search

Captured 2026-07-26 using the required grouped term `white-blood-cell`. The result page reports about 125 results, compared with about 491,000,000 for `elephant` under the same settings.

Open the reproducible Google Search query

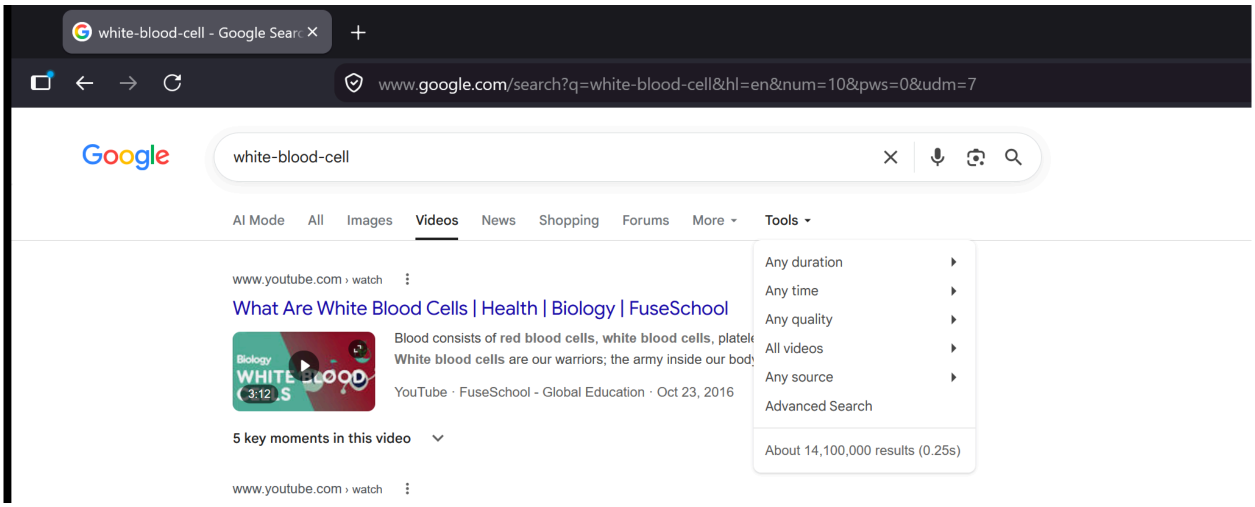


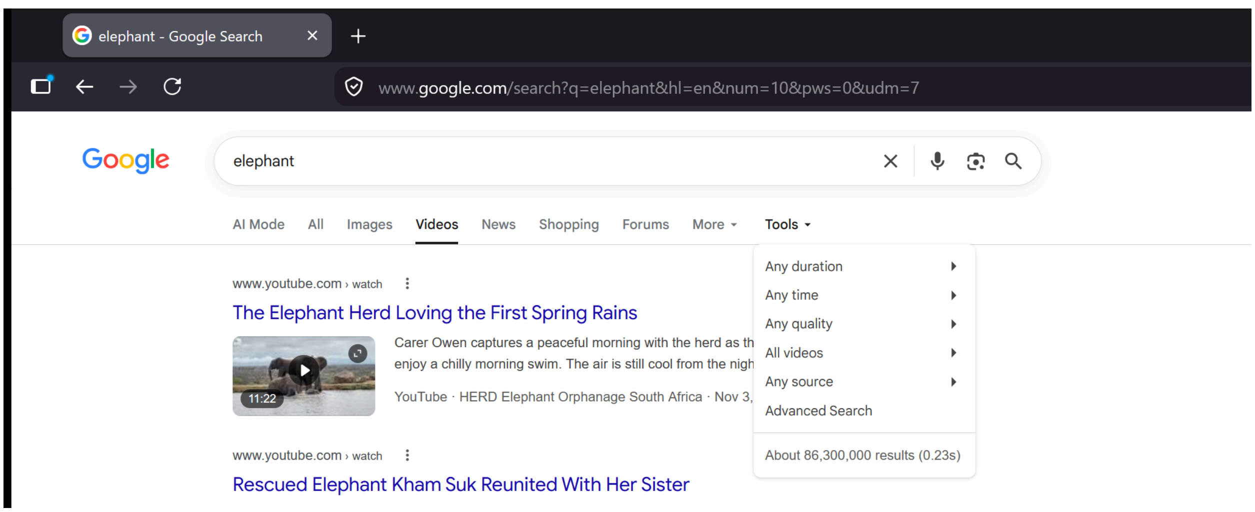


Google Video Search

Captured 2026-07-26 using the required grouped term `white-blood-cell`. The video result page reports about 14,100,000 results, compared with about 86,300,000 for `elephant` under the same settings.

Open the reproducible Google Video Search query

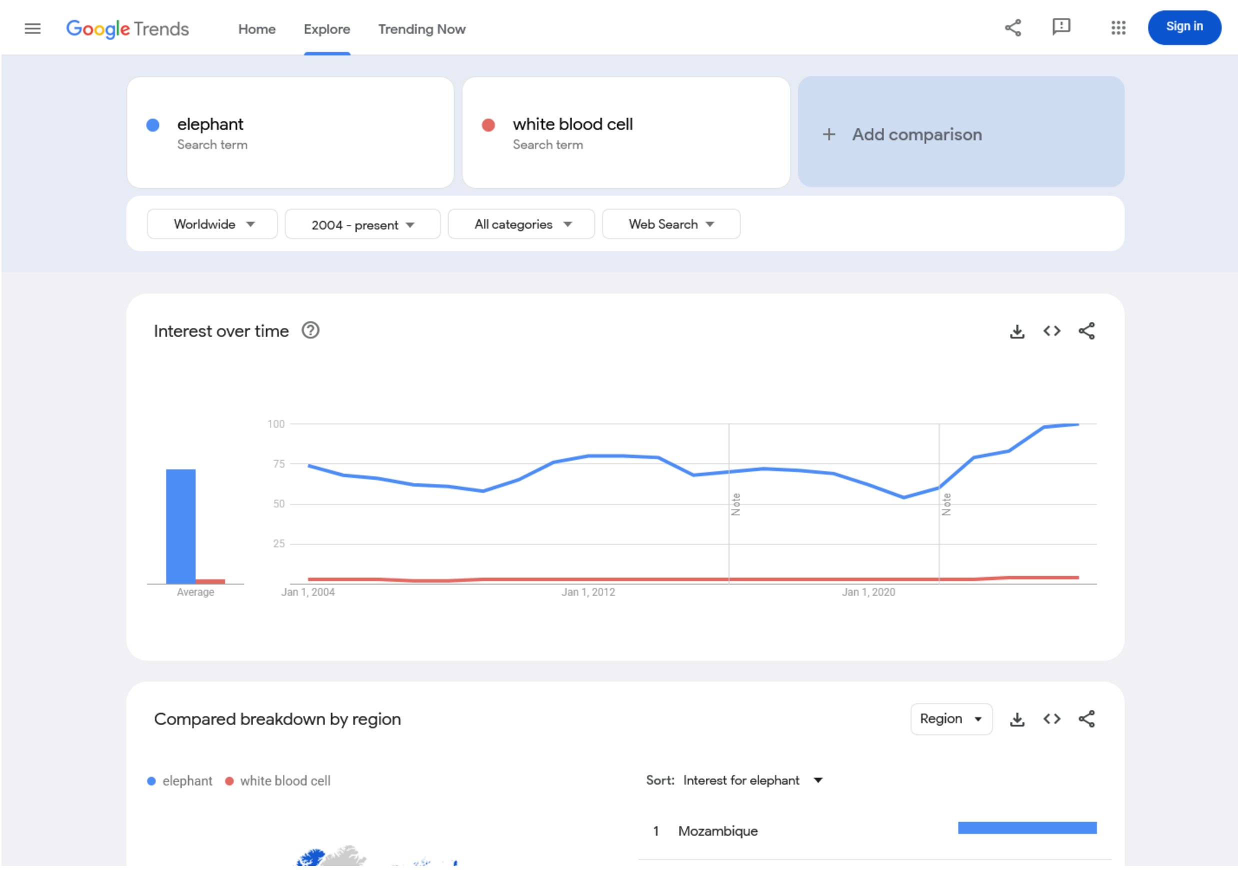




Google Trends - Web Search

Captured 2026-07-26 using Worldwide, 2004-present, all categories, and Web Search. Average indexed interest was 72 for `elephant` and 3 for `white blood cell`; the white-blood-cell series reaches 4 in 2024, 2025, and 2026.

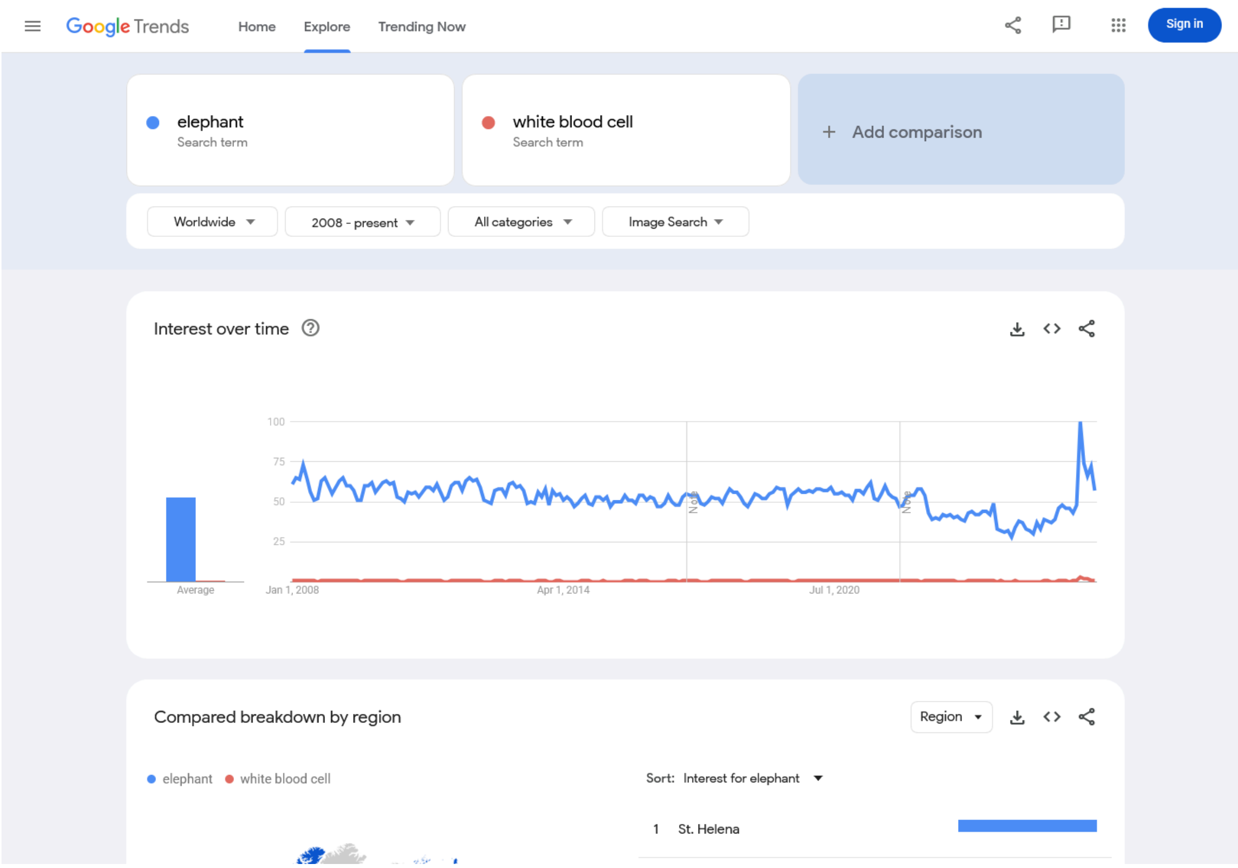
Open the Google Trends Web Search comparison



Google Trends - Image Search

Captured 2026-07-26 using Worldwide, 2008-present, all categories, and Image Search. Average indexed interest was 53 for `elephant` and 1 for `white blood cell`, with recurring white-blood-cell interest across the period.

Open the Google Trends Image Search comparison



Google Books Ngram Viewer

Captured 2026-07-26 using the English corpus, 1500-2022, and smoothing 3. In 2022, the frequency of `white blood cell` was approximately 10.13% of `elephant`, with an increasing modern series.

[Open the Google Books Ngram comparison]